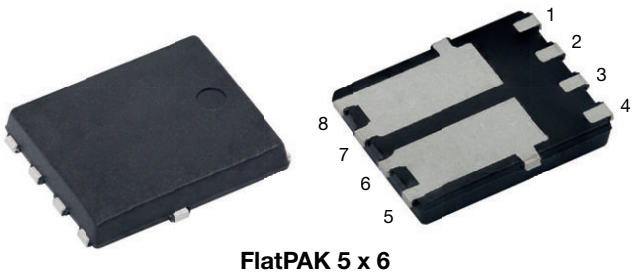
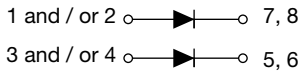
High Current Density Surface-Mount Trench MOS Barrier Schottky Rectifier

Ultra Low V_F = 0.44 V at I_F = 1.5 A



FlatPAK 5 x 6



LINKS TO ADDITIONAL RESOURCES

3D Models | Packages | Marking

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	2 x 3 A
V_{RRM}	100 V
I_{FSM}	80 A
V_F at I_F = 3 A (T_A = 125 °C)	0.55 V
T_J max.	150 °C
Package	FlatPAK 5 x 6
Circuit configuration	Separated cathode

FEATURES

AUTOMOTIVE GRADE Available

RoHS COMPLIANT

HALOGEN FREE

- Trench MOS Schottky technology
- Low forward voltage drop, low power losses
- High efficiency operation
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- AEC-Q101 qualified available
  - Automotive ordering code: base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

TYPICAL APPLICATIONS

For use in low voltage high frequency DC/DC converters, freewheeling diodes, and polarity protection applications.

MECHANICAL DATA

Case: FlatPAK 5 x 6

Molding compound meets UL 94 V-0 flammability rating

Base P/N-M3 - halogen-free, RoHS-compliant

Base P/NHM3 - halogen-free, RoHS-compliant, and AEC-Q101 qualified

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102

M3 and HM3 suffix meets JESD 201 class 2 whisker test

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)			
PARAMETER	**SYMBOL**	**V6K100DU**	**UNIT**
Device marking code		V610D	
Maximum repetitive peak reverse voltage	V_{RRM}	100	V
Maximum DC forward current per diode	$I_{F(AV)}$ (1)	3	A
	$I_{F(AV)}$ (2)	2.2	A
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load per diode	I_{FSM}	80	A
Operating junction temperature range	T_J (3)	-40 to +150	°C
Storage temperature range	T_{STG}	-55 to +150	°C

Notes

(1) With infinite heatsink

(2) Free air, mounted on recommended pad area

(3) The heat generated must be less than the thermal conductivity from junction-to-ambient: $dP_D/dT_J < 1/R_{\theta JA}$

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Instantaneous forward voltage per diode	I_F = 1.5 A	T_A = 25 °C	V_F (1)	0.50	-	V
	I_F = 3 A			0.61	0.69	
	I_F = 1.5 A	T_A = 125 °C		0.44	-	
	I_F = 3 A			0.55	0.63	
Reverse current per diode	V_R = 70 V	T_A = 25 °C	I_R (2)	0.01	-	mA
		T_A = 125 °C		2	-	
	V_R = 100 V	T_A = 25 °C		-	0.35	
		T_A = 125 °C		4	12	
Typical junction capacitance per diode	4.0 V, 1 MHz		C_J	360	-	pF

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: pulse width ≤ 5 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)				
PARAMETER	**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Thermal resistance per diode	$R_{\theta JA}$ (1)(2)	100	-	°C/W
	$R_{\theta JM}$ (3)	3.5	4.5	

Notes

(1) The heat generated must be less than thermal conductivity from junction-to-ambient: $dP_D/dT_J < 1/R_{\theta JA}$

(2) Free air, mounted on recommended copper pad area; thermal resistance $R_{\theta JA}$ - junction-to-ambient

(3) Mounted on infinite heat sink; thermal resistance $R_{\theta JM}$ - junction-to-mount

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PREFERRED PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
V6K100DU-M3/H	0.10	H	1500	7" diameter plastic tape and reel
V6K100DU-M3/I	0.10	I	6000	13" diameter plastic tape and reel
V6K100DU-M3/IE (2)	0.10	IE	6000	13" diameter plastic tape and reel
V6K100DUHM3/H (1)	0.10	H	1500	7" diameter plastic tape and reel
V6K100DUHM3/I (1)	0.10	I	6000	13" diameter plastic tape and reel
V6K100DUHM3/IE (1)(2)	0.10	IE	6000	13" diameter plastic tape and reel

Notes

(1) AEC-Q101 qualified

(2) Component rotate 90° in cavity

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

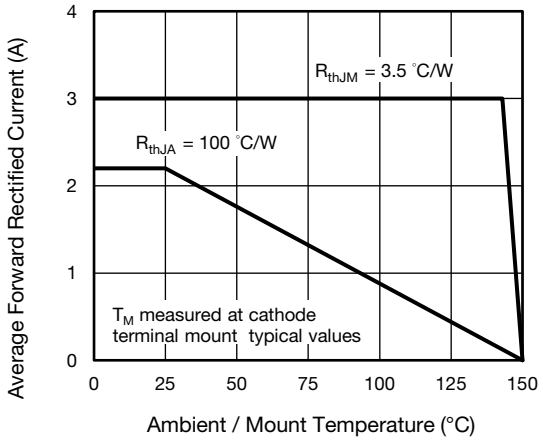


Fig. 1 - Maximum Forward Current Derating Curve

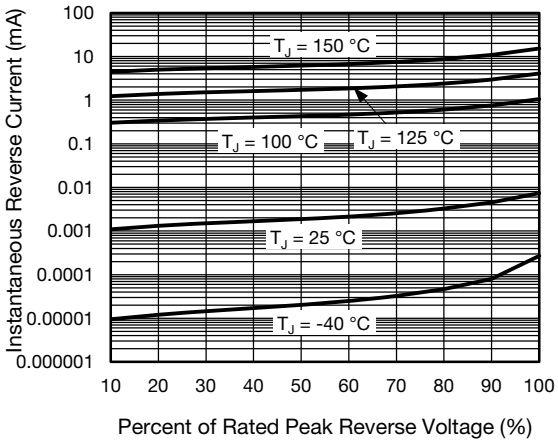


Fig. 4 - Typical Reverse Leakage Characteristics

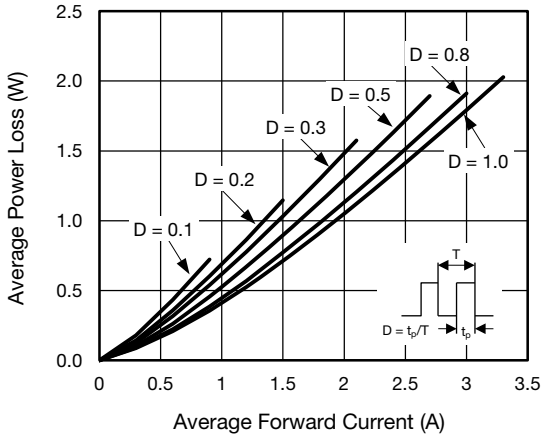


Fig. 2 - Forward Power Loss Characteristics

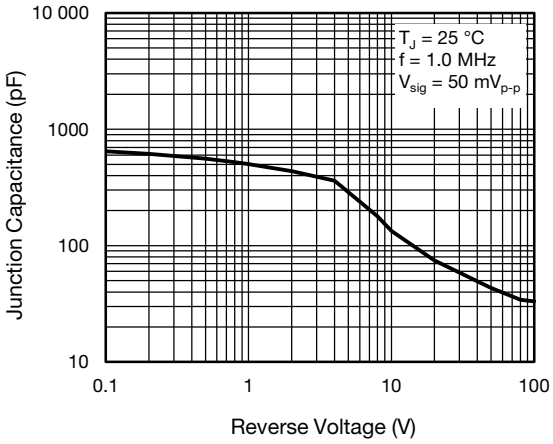


Fig. 5 - Typical Junction Capacitance

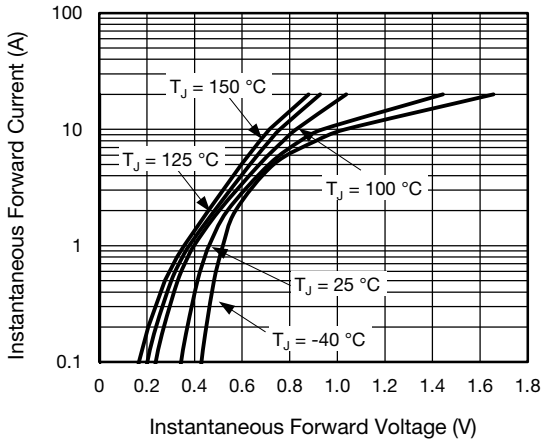


Fig. 3 - Typical Instantaneous Forward Characteristics

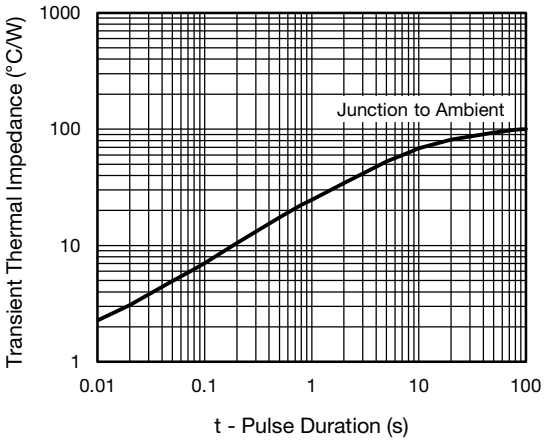
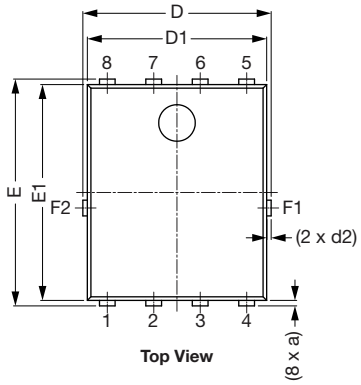


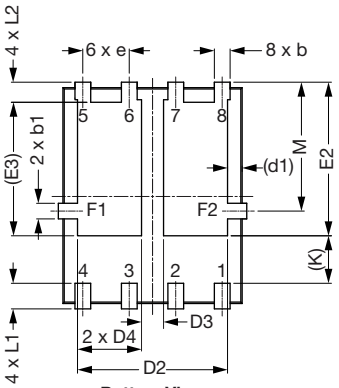
Fig. 6 - Typical Transient Thermal Impedance

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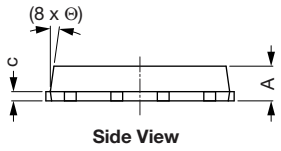
PACKAGE OUTLINE DIMENSIONS in inches (millimeters)



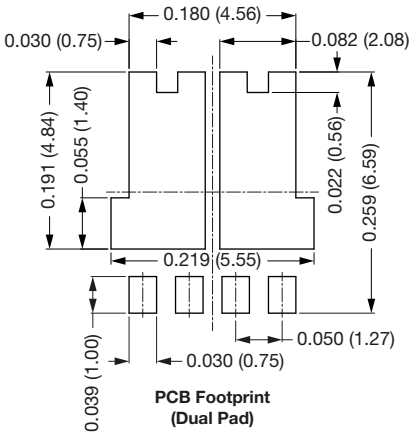
Top View



Bottom View
(Dual Pad)



Side View



PCB Footprint
(Dual Pad)

DIM.	INCHES			MILLIMETERS		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	0.035	0.039	0.043	0.89	0.99	1.09
(a)	-	0.006	-	-	0.15	-
b	0.013	0.017	0.020	0.32	0.43	0.52
b1	0.013	0.017	0.020	0.32	0.43	0.52
c	0.008	-	0.014	0.20	-	0.35
D	0.197	0.203	0.209	5.00	5.15	5.30
D1	0.189	0.193	0.197	4.80	4.90	5.00
D2	0.154	0.161	0.169	3.90	4.10	4.30
D3	0.020	0.024	0.031	0.50	0.60	0.80
D4	0.063	0.069	0.075	1.60	1.75	1.90
(d1)	-	0.016	-	-	0.40	-
(d2)	-	0.005	-	-	0.125	-
E	0.238	0.244	0.250	6.05	6.20	6.35
E1	0.228	0.232	0.236	5.80	5.90	6.00
E2	0.157	0.165	0.173	4.00	4.20	4.40
(E3)	-	0.144	-	-	3.65	-
e	0.050 BSC			1.27 BSC		
(K)	0.039	-	-	1.00	-	-
L1	0.019	-	0.043	0.48	-	1.10
L2	0.012	-	0.031	0.30	-	0.80
M	0.128	0.138	0.148	3.25	3.50	3.75
Θ	0°	-	10°	0°	-	10°

Notes

- Dimensioning and tolerancing per ASME Y14.5-2009
- Dimensions D1 and E1 do not include mold flash or gate burrs
- Dimension (XX) means reference only

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